WORKED SOLUTION

STEP II 2013, Question 8

Pure • Calculus and Inequalities

STEP II 2013, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s2-q8

The key idea

Because f is decreasing, the region is a "staircase" that only gets lower as we move right, so the best rectangle sits in the bottom-left corner, on the line $y = \mathrm{f}(t)$ and against the y-axis, with its top right corner on the curve. That reduces part (i) to maximising a function of one variable. Part (ii) compares two areas, the whole region and the rectangle inside it, and part (iii) simply makes everything explicit for $\mathrm{f}(x) = 1/(1+x)$.

Solution

Part (i)

The region is $\{(x,y) : 0 \le x \le t,\ \mathrm{f}(t) \le y \le \mathrm{f}(x)\}$: the curve lies above the line $y = \mathrm{f}(t)$ for $x < t$ since f is decreasing.

Take any rectangle $[a,b] \times [c,d]$ inside the region. Its bottom edge must have $c \ge \mathrm{f}(t)$, its left edge $a \ge 0$, and its top right corner (b,d) must lie in the region, so $b \le t$ and $d \le \mathrm{f}(b)$. Hence its area is at most

$$b(\mathrm{f}(b) - \mathrm{f}(t)),$$

and this is achieved by the rectangle $[0,b] \times [\mathrm{f}(t), \mathrm{f}(b)]$, which does fit: for $0 \le x \le b$ we have $\mathrm{f}(x) \ge \mathrm{f}(b)$, so the rectangle lies under the curve. So

$$A_0(t) = \max_{0 \le x \le t} R(x), \qquad R(x) = x(\mathrm{f}(x) - \mathrm{f}(t)).$$

Now $R(0) = R(t) = 0$ while $R(x) > 0$ for $0 < x < t$, so for $t > 0$ the maximum is attained at an interior point $x_0 \in (0,t)$, where $R'(x_0) = 0$:

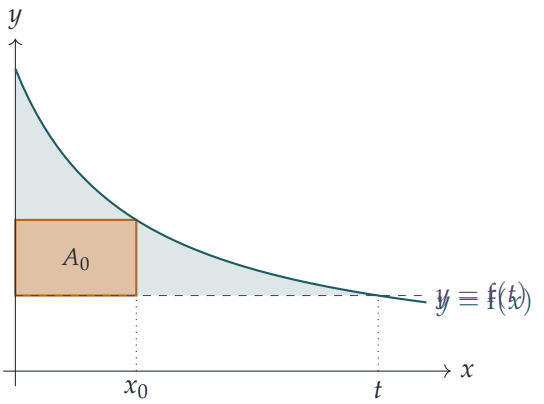
$$R'(x) = \mathrm{f}(x) + x\mathrm{f}'(x) - \mathrm{f}(t) = 0 \quad \Longrightarrow \quad x_0\mathrm{f}'(x_0) + \mathrm{f}(x_0) = \mathrm{f}(t).$$

Thus $A_0(t) = x_0(\mathrm{f}(x_0) - \mathrm{f}(t))$ with x_0 satisfying this condition, as required. (For $t = 0$ the region has no width, $A_0(0) = 0$, and $x_0 = 0$ satisfies the condition trivially.)

Part (ii)

Write $t\,g(t) = \int_0^t f(x)\,dx$. Differentiating both sides, using the product rule on the left and the fundamental theorem of calculus on the right,

$$g(t) + t\,g'(t) = f(t) \quad \Longrightarrow \quad t\,g'(t) = f(t) - g(t).$$



The integral $\int_0^t \big(f(x) - f(t)\big)\,dx$ is exactly the area of the shaded region, between the curve and the line $y = f(t)$ for $0 \le x \le t$. The optimal rectangle from part (i) lies inside this region, and it does not fill it: for $x_0 < x < t$ the rectangle contributes nothing, but the region has height $f(x) - f(t) > 0$ there, since f is strictly decreasing. So that sliver has positive area, and

$$\int_0^t \big(f(x) - f(t)\big)\,dx > A_0(t).$$

? **Ask yourself** How do we turn the left-hand side into something involving g'?

Split the integral and use the definition of g, then the result just proved:

$$\begin{aligned}\int_0^t \big(f(x) - f(t)\big)\,dx &= t\,g(t) - t\,f(t)\\ &= -t\big(f(t) - g(t)\big)\\ &= -t\cdot t\,g'(t) = -t^2 g'(t).\end{aligned}$$

Hence $-t^2 g'(t) > A_0(t)$.

Part (iii)

Here $f(x) = \dfrac{1}{1+x}$ is positive and strictly decreasing for $x \ge 0$, so everything above applies.

The left-hand side. From the calculation at the end of part (ii),

$$-t^2 g'(t) = \int_0^t \frac{dx}{1+x} - \frac{t}{1+t} = \ln(1+t) - \frac{t}{1+t}.$$

The rectangle. With $\mathrm{f}'(x) = -\dfrac{1}{(1+x)^2}$,

$$x\mathrm{f}'(x) + \mathrm{f}(x) = \frac{-x + (1+x)}{(1+x)^2} = \frac{1}{(1+x)^2},$$

so the condition for x_0 is $(1+x_0)^2 = 1+t$. Write $s = \sqrt{1+t}$, so $s > 1$ and $x_0 = s-1$ (the positive root; note $0 < x_0 < t$). Then $1+t = s^2$ and

$$A_0(t) = (s-1)\left(\frac{1}{s} - \frac{1}{s^2}\right) = \frac{(s-1)^2}{s^2} = \left(1 - \frac{1}{s}\right)^2.$$

Ask yourself The two sides look messy: what is the natural variable?

Everything is cleaner in terms of s. We have $\ln(1+t) = 2\ln s$ and $\dfrac{t}{1+t} = 1 - \dfrac{1}{s^2}$, so $-t^2\mathrm{g}'(t) > A_0(t)$ becomes

$$\begin{aligned} 2\ln s - 1 + \frac{1}{s^2} &> 1 - \frac{2}{s} + \frac{1}{s^2} \\ \iff \quad 2\ln s &> 2 - \frac{2}{s} \\ \iff \quad \ln s &> 1 - \frac{1}{s}. \end{aligned}$$

Since $s = \sqrt{1+t}$ and $\ln s = \ln\sqrt{1+t}$, this is

$$\ln\sqrt{1+t} > 1 - \frac{1}{\sqrt{1+t}} \qquad (t > 0),$$

as required.

Checks

- With $t = 3$: $s = 2$, $x_0 = 1$, $A_0 = \frac{1}{4}$, and the region has area $\ln 4 - \frac{3}{4} \approx 0.636 > 0.25$. The final inequality reads $\ln 2 \approx 0.693 > 0.5$.
- A direct search over rectangles $x\big(\mathrm{f}(x) - \mathrm{f}(t)\big)$ for many values of t confirmed, by computer, that the maximum is $(1 - 1/\sqrt{1+t})^2$, attained at $x = \sqrt{1+t} - 1$.
- As $t \to 0^+$, both sides of the final inequality tend to 0; expanding, $\ln s \approx (s-1) - \frac{1}{2}(s-1)^2$ and $1 - \frac{1}{s} \approx (s-1) - (s-1)^2$, so the left side wins by about $\frac{1}{2}(s-1)^2$, as it should.

Common slips

- Assuming without comment that the best rectangle has its corner on the curve: it needs the decreasing property, as in part (i).

- Forgetting to say why the area inequality is *strict*: the rectangle misses the part of the region with $x_0 < x < t$.
- Taking the negative root $1 + x_0 = -\sqrt{1+t}$, which gives $x_0 < 0$.

Beyond STEP

The final inequality, $\ln s > 1 - 1/s$ for $s > 1$, is one half of the classic bounds $1 - \frac{1}{s} < \ln s < s - 1$, which come from comparing $\int_1^s \mathrm{d}u/u$ with rectangles; this question reaches it by a cleverer geometric route. Comparing integrals with areas of rectangles is the heart of the Riemann integral, which first-year analysis courses build carefully from upper and lower sums. Abbott's *Understanding Analysis* gives a very readable account.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.